Evolution and Human Behavior 46 (2025) 106749

Contents lists available at ScienceDirect

Evolution and Human Behavior

journal homepage: www.elsevier.com/locate/ens

Review Article

Replication of experiments and the canonisation of incorrect conclusions

Stuart A. West [a,*], Maxwell N. Burton-Chellew [a,b]

[a] Department of Biology, University of Oxford, Oxford OX1 3RB, UK
[b] Department of Economics, University of Lausanne, CH-1015 Lausanne, Switzerland

ARTICLE INFO

Keywords:
Controls
Cooperation
Public goods
Null hypothesis
Emperor's new clothes

ABSTRACT

We are in the middle of a revolution over how to collect, analyse and interpret results in the human behavioural sciences. A key issue is to determine the extent to which experimental results are repeatable. However, if the experiment being replicated lacks the appropriate controls or null hypotheses, then replication of experiments can lead to the canonisation of incorrect conclusions. Experiments are required that allow a strong possibility to falsify hypotheses or allow competing hypotheses to be tested in an unbiased manner. We illustrate this problem, and how it can be solved, with examples from the literature on public goods games, which has been used to test hypotheses about the evolution of cooperation in humans.

1. Introduction

A fundamental feature of good scientific practise is the extent to which results can be replicated (Fisher, 1935; Popper, 1959). The more that an experimental result can be shown to be independently repeatable, and that there is not a publication bias towards results in a certain direction, the more that result can be trusted (Borenstein, Hedges, Higgins, & Rothstein, 2021; Camerer et al., 2018; Fletcher, 2021; Nissen, Magidson, Gross, & Bergstrom, 2016; Nosek et al., 2022; Nosek & Errington, 2020; Palmer, 2000). However, the conclusions that can be drawn from repeatable experiments depend upon the design of the experiments used to replicate (Bergstrom & West, 2021; Derksen & Morawski, 2022; Fisher, 1935; Hedges & Schauer, 2021; Locey, 2020; Nosek & Errington, 2020; Ruxton & Colegrave, 2016). If an experiment lacks an appropriate control or null hypothesis, or does not allow a hypothesis to be falsified, then its repetition could make acceptance of a misleading conclusion more rather than less likely (Fig. 1).

To give a hypothetical and deliberately simple example, suppose we hypothesised that putting water into a drinking vessel turns it red. We then took 30 mugs made with red ochre and added water to them, and found that a significant number turned the water red. Then we repeated this exact experiment 100 times and kept getting the same result. Although we have a very repeatable result, we would be foolish to claim strong support for our hypothesis that putting water into a drinking vessel turns it red. We have not varied the type of drinking vessel, or what it was made from, or offered competing hypotheses, or made a strong attempt to falsify our hypothesis.

We suggest that an analogous problem can occur in the human behavioural sciences. Repetition of experiments that lack appropriate controls or null hypotheses, or do not allow a hypothesis to be easily falsified, can lead to canonisation of misleading conclusions. Consequently, although it could appear that an attempt to test a hypothesis is being replicated, the test is essentially being pseudoreplicated. If the test is biased, then this can lead to the canonization of false conclusions. Then are several different ways to look at this problem, which we shall return to below – it is an example of the need for both replication that can provide diagnostic evidence for or against a previous claim, and experiments based on testable / falsifiable theories(Nosek & Errington, 2020). We distinguish here from work that has demonstrated the problem of methodologies that lead to false positive results – our emphasis is when methodologies are biased against, or even do not allow, negative results (Borenstein et al., 2021; Gelman & Loken, 2013, 2015; Head, Holman, Lanfear, Kahn, & Jennions, 2015; Ioannidis, 2005; Nissen et al., 2016; Palmer, 2000; Sefton & Steinberg, 1996; Simmons, Nelson, & Simonsohn, 2011).

We illustrate this problem with examples from the literature on behaviour in public goods games. Economic games such as public goods games have frequently been used to test hypotheses about both the proximate and ultimate factors leading to cooperation in humans (Fehr & Schurtenberger, 2018). We focus on this case because: (1) it provides a clear example of the problem discussed above; and (2) the potential problems have been raised multiple times, and so we can examine how the field has responded to this. Our aim is to raise issues about the scientific method rather than to debate hypotheses.

* Corresponding author.
E-mail address: stuart.west@biology.ox.ac.uk (S.A. West).

https://doi.org/10.1016/j.evolhumbehav.2025.106749
Received 6 March 2025; Accepted 11 August 2025
Available online 19 August 2025
1090-5138/© 2025 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

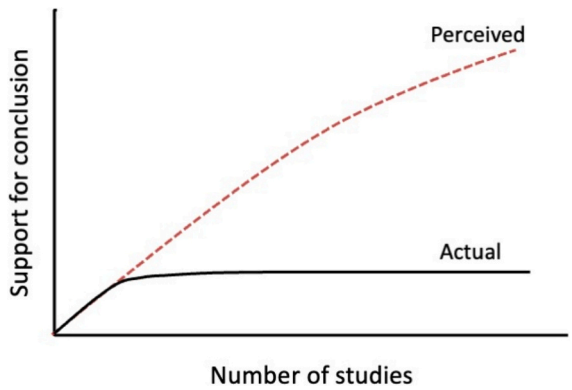

Fig. 1. Canonisation of untested theory. As an experiment is replicated, this can lead to increased support for (or belief in) a conclusion that is consistent with the data. However, if the data is consistent with an alternative competing hypothesis, or multiple competing hypotheses, then there will be far less actual support for the conclusion that is becoming accepted.

2. Public goods games

The public goods game is an economic experiment that is used to study cooperation in humans (Ledyard, 1995). In this game, individuals are placed into groups of size N and are then endowed with some money that they can choose to contribute to a group fund (public good). Any money contributed is multiplied by an amount M (where $M > 1$) and then shared equally between the members of the group. In most games, and the scenario that we shall focus on, the money is multiplied by less than the number of players ($M < N$), and so the return to an individual attributable from their contribution will be less than they contributed.

Consequently, although the group does best if everyone cooperates by contributing fully (because $M > 1$), an individual can always gain a higher income from a single interaction by not contributing (because $M < N$). Individuals can be favoured to contribute when playing alternate versions of the game – for example, if there is the chance for repeated interactions with the same individuals, and hence reciprocal cooperation, or if non-contributors are punished or ostracized (Fehr & Gächter, 2002; Zelmer, 2003). We shall focus, for simplicity, on the simple scenario, where the game is repeatedly played anonymously with strangers and there is no punishment.

3. Results and interpretation

The standard result in public goods games is that when groups of inexperienced players play the game, they start by contributing about 50 % of their endowment (Ledyard, 1995). Then, as the game is repeated, with fresh endowments each round, contributions decline, to about 20 % of the endowments. This experiment has been repeated, with similar results, hundreds of times (Burton-Chellew & West, 2021; Fehr & Gächter, 2002; Ledyard, 1995; Thöni & Volk, 2018; Zelmer, 2003) (Fig. 2).

The most common interpretation of this result is that most, but not all, humans are altruistic and have 'prosocial preferences' which induce them to cooperate with unrelated individuals, even when this is costly and provides no benefit (sometimes termed 'conditional cooperators') (Fehr & Fischbacher, 2003; Gintis, Bowles, Boyd, & Fehr, 2003; Henrich et al., 2005; Weber, Weisel, & Gächter, 2018). Cooperation is suggested to reduce (unravel) because the presence of a minority of stubborn selfish individuals leads to frustration among altruists who gradually withdraw their cooperation in response. Papers on this topic often take this interpretation as their accepted starting point, from which more elaborate details can be examined, rather than treating it as one possible hypothesis.

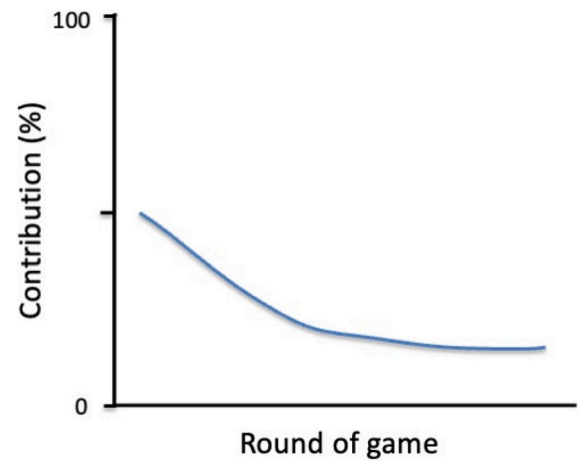

Fig. 2. The typical result in public goods games experiments. Across hundreds of studies, contributions (cooperation) begin around 40–50 % and then steadily declines in later rounds (Burton-Chellew & West, 2021; Ledyard, 1995).

4. Competing hypotheses

A potential problem with the interpretation of public goods games given above is the extent to which the prosocial preferences hypothesis can be falsified. Individuals can contribute 0–100 % of their money. The contribution that would maximise their return is 0 %. Consequently, in this experimental design: (i) individuals are unable to contribute less than would maximise their income (<0 % is not possible); (ii) any contribution, except the single case of contributing exactly 0 %, would lead to individuals behaving in a costly way that benefited others (Burton-Chellew & West, 2013; Kümmerli, Burton-Chellew, Ross-Gillespie, & West, 2010). Any form of contributions would appear altruistic and so there is a bias towards finding this result (i.e. the null hypothesis of exactly 0 % contribution is too easy to reject). If researchers assume that the behaviour of individuals perfectly reflects their preferences, then any contribution for any reason would lead to the assumption that individuals will appear to be incurring costs *in order to* provide benefits to others.

However, care needs to be taken when linking observations to cause – as Adam Smith pointed out, just because the baker provides others with food, that is not the cause of his behaviour. It is possible to come up with a range of hypotheses for why individuals could contribute something to the group fund (Andreoni, 1995a; Andreoni, 1995b; Bateson, Nettle, & Roberts, 2006; Binmore & Shaked, 2010; Burton-Chellew, Mouden, & West, 2016; Burton-Chellew & West, 2013; Ferraro & Vossler, 2017; Hagen & Hammerstein, 2006; Haley & Fessler, 2005; Henrich et al., 2005; Houser & Kurzban, 2008; Kümmerli et al., 2010) (i.e. >0 %) (Fig. 3). These include:

- Individuals could map the game to scenarios that they encounter in the real world, where factors such as reciprocity or reputation might be important, rather than the exact scenario envisaged by the experimenter.
- Individuals might have an aversion to extreme behaviours such as 0 % or 100 % contribution and prefer to 'bet-hedge' by contributing something in-between these extremes.
- Individuals may misunderstand and be confused about the incentives, but then learn from experience how to improve their earnings.
- Individuals might be susceptible to suggestions from the experimenter's instructions on how to behave.
- Individuals might be prone to playing randomly.
- Individuals might want to explore all the behavioural options the experimenter makes available to them - meaning that in the space of

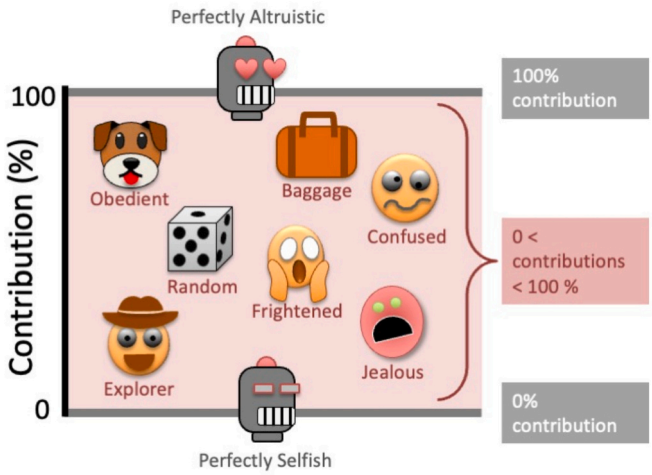

Fig. 3. Biased designs can favour prosocial interpretations. If players express their preferences perfectly then selfish players will only ever contribute 0 % ('Perfectly Selfish'), meaning that all contributions >0 % have been interpreted as prosocial. Some patterns of contribution above 0 % are consistent with prosocial preferences. Fully altruistic players would only ever contribute 100 % ('Perfectly Altruistic'), and conditional cooperators would attempt to match the cooperation of others and contribute intermediate amounts ('Jealous'). However, there are many reasons for above zero contributions. For example, players may: (i) play randomly ('Random'); (ii) like to explore their options in experiments ('Explorer'); (iii) be risk averse or uncertain avoiding extreme behaviours such as close to 0 or 100 % ('Frightened'); (iv) be confused about the game's payoffs and make mistakes that fail to express their true preferences ('Confused'); (v) import behaviours from the real world (cultural 'Baggage'); (vi) be trying to behave in ways that they think will please the experimenter ('Obedient').

an hour long experiment they will want to try different behaviours even if they suspect them to be costly.

These factors could be especially relevant when testing individuals who are not experienced in economic games.

Our major point is not to argue which hypothesis best explains the data (which doesn't matter for our purpose here), but rather that repetition of a core, biased, experiment is insufficient because there is a need to design experiments that explicitly test between competing hypotheses. This is not just a debate about different theories or hypotheses – it cuts to the core of the scientific method. Unfortunately, a common course of action has been to repeat the standard public goods experiment and assume that the data demonstrate prosocial preferences (Andreoni, 1995a; Bateson et al., 2006; Burnham & Hare, 2007; Burton-Chellew et al., 2016; Burton-Chellew, Nax, & West, 2015; Burton-Chellew & West, 2013; Fehr & Charness, 2023; Fehr & Fischbacher, 2003; Fehr & Schurtenberger, 2018; Ferraro & Vossler, 2017; Gächter & Fages, 2023; Houser & Kurzban, 2008; Kümmerli et al., 2010; Shapiro, 2008). We are not saying that everyone accepts this, but just that it is done sufficiently that there is a fundamental methodological problem.

5. Alternative perspectives

There are alternate ways of conceptualising or describing the problem that we have given above. One alternative is that the standard public good game involves no control for the prosocial preferences hypothesis. A control would be required where we could study behaviour in a situation with no potential for prosocial preferences, or where there is also potential for antisocial preferences. Another alternative is that a more appropriate, behaviourally valid, null hypothesis is required. In the standard game, only the single case of 0 % contribution would support the null hypothesis, and so almost the entire range of the parameter space (>0 %) would support the prosocial hypothesis. Treatments are needed that allow us to infer that the financial consequences of decisions that are psychologically salient.

We have focused on the issue of experimental design, and the need for experiments that have a fair chance to falsify a theory. This point has been made more generally by Nosek & Errington, who define a replicate based on its ability to provide "diagnostic evidence about a claim from prior research" (Nosek & Errington, 2020). By these criteria, the numerous repeats of the public goods game would not count as replicate tests of the prosocial preferences hypothesis. We are not suggesting that all the experiments were carried out with this purpose, but rather that the repeated result did not provide additional support. The problem can also be looked at from the perspective of the underlying theory – we need to articulate theory such that it can be "productively confronted with evidence" (Nosek & Errington, 2020). In our hypothetical mug example, it was the flawed theory of mugs that led to the over-generalisation from water in mugs experiments. In the public goods game, it was the underlying theoretical hypothesis that led to conclusions about what could be learnt from simple public good games. We need to replicate good experiments based on good theories.

6. Controls, null hypotheses, and falsification in practise

We could just stop here, as our main purpose is to just make the point that controls, null hypotheses and competing hypotheses are required. It doesn't matter for the purpose of this paper what the final explanation for behaviour in public goods games is. What matters is that we need to get to the right answer by a rigorous a scientific method as possible. However, the potential problems have been raised with respect to public goods games, and attempts have been made to test competing hypotheses (Andreoni, 1995b; Binmore & Shaked, 2010; Burton-Chellew, 2022; Burton-Chellew et al., 2016; Burton-Chellew, D'Amico, & Guérin, 2022; Burton-Chellew & Guérin, 2022; Burton-Chellew, Mouden, & West, 2017; Burton-Chellew & West, 2013, 2021, 2022; Haley & Fessler, 2005; Kümmerli et al., 2010; Nax, Burton-Chellew, West, & Young, 2016). Consequently, this allows us to examine how different solutions to the problem can be developed, the utility of different methodologies, and the response to these issues.

6.1. *Option 1: Control treatments that allow behaviour to go in both directions*

One option is to develop controls for possible causes of experimental bias. We discussed above how the potential results of the usual public good game are biased in one direction. It is only possible to contribute more than would maximise personal gain (>0 %), and not less (<0 % is not possible). A way to make the opposite pattern possible is to modify the public good game into a 'public delight' game, by making the multiplication factor greater than the number of players ($M > N$) (Kümmerli et al., 2010). In this case, the behaviour that would maximise the earnings of the individual is 100 % contribution, which is also the behaviour that maximises earnings for the other members of the group. The public delight game aligns self-interested and prosocial preferences.

When individuals play public delight games, where $M > N$, they do not contribute 100 % (Fig. 4a). Instead, individuals behave in a way that is near symmetrical to the standard public goods game, but in the opposite direction. Individuals start by contributing about 50 %, and then contributions increase over time, but do not reach 100 % (Kümmerli et al., 2010). Consequently, individuals are contributing less than would maximise their own income, but this also imposes a cost on the income of others. If we assumed that behaviour was completely dependent upon preferences, then we would need to invoke antisocial preferences to explain the data. This conclusion is incompatible with standard 'pro-social' conclusions from the public good game. A more parsimonious conclusion from data collected in public good and public delight games is that we should reject the hypothesis that individuals are playing 'perfectly'.

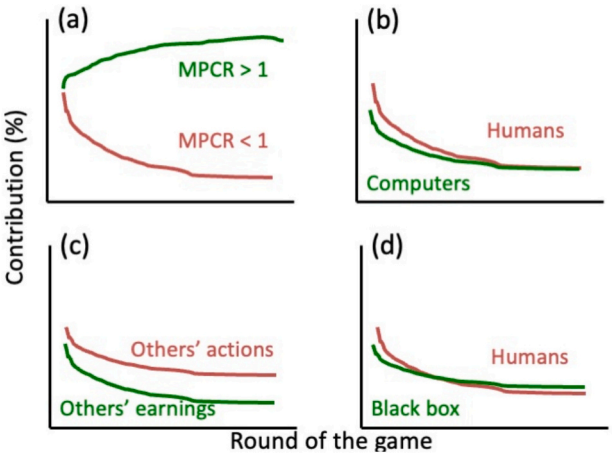

Fig. 4. Typical result (red data) and the results from different types of control (green data). (a) When the individual benefit to contributing is positive rather than negative (MPCR>1), both selfish and prosocial individuals should contribute (cooperate) fully, but they do not. Instead, contributions start at intermediate values and slowly increases over time (Kümmerli et al., 2010). (b) The level of contribution is not altered by whether individuals are playing with either humans or computers (for which they could have no prosocial preferences)(Burton-Chellew, 2022; Burton-Chellew et al., 2016; Burton-Chellew et al., 2022; Ferraro & Vossler, 2017; Houser & Kurzban, 2008; Shapiro, 2008). (c) When individuals observe the earnings of other players, contributions decline more rapidly(Burton-Chellew et al., 2017; Burton-Chellew & D'Amico, 2021). (d) People play the same way even when they do not know they are playing a public good game, and instead think they are trying to optimize how many coins to input into a 'black box' (where there can be no prosocial preferences)(Burton-Chellew & West, 2013, 2022). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

6.2. Option 2: Asocial controls that remove social preferences

Another option is to develop controls where there is no possibility for prosocial preferences. One way is to ask individuals to play the same game but with computers, who cannot benefit from their actions (Burton-Chellew, 2022; Burton-Chellew et al., 2016; Burton-Chellew et al., 2022; Ferraro & Vossler, 2017; Houser & Kurzban, 2008; Shapiro, 2008). Yet another option is to make individuals play a public goods game but not tell them they are in one – and instead tell them that can put money into a virtual 'black box', which will then return some output, determined by a hidden mathematical function (Burton-Chellew et al., 2015; Burton-Chellew & West, 2013, 2022; Nax et al., 2016). The shared feature of these two options is that they remove any social aspect to the game (in different ways) and hence measure how a population of uncertain players with no prosocial preferences behaves.

With both these 'asocial' controls, individuals showed patterns of behaviour that were essentially the same as when knowingly playing a public goods game with other humans (Burton-Chellew et al., 2016; Burton-Chellew & West, 2013, 2022). When playing with computers, players still based their behaviour on the actions of their groupmates (Fig. 4b). When playing into a black box, players learned over time to contribute less (Fig. 4d). Consequently, prosocial preferences are not required to obtain the standard patterns of behaviour. The usefulness of such control treatments is that they provide a more behaviourally valid null hypothesis than the extreme predictions of a perfectly behaved selfish agent (*Homo economicus*).

6.3. Option 3: Testing for perfect play

Another option is to directly test whether individuals play games 'perfectly'. If people play economic games perfectly then providing them with extra, but redundant, information should not affect their behaviour. However, when players are given enhanced information during games that makes the costs of contributing clearer, they more quickly stop cooperating (Burton-Chellew et al., 2015; Burton-Chellew & Guérin, 2022; Burton-Chellew & West, 2013). For example, an increased awareness of how cooperation benefits others leads to a reduction, rather than an increase, in the level of cooperation (Fig. 3c) (Burton-Chellew et al., 2015; Burton-Chellew & Guérin, 2022; Burton-Chellew & West, 2013). This is consistent with the hypothesis that individuals are confused about the potential costs and benefits of contributing.

6.4. Option 4: Explicitly testing competing hypotheses

Another possibility is to develop competing hypotheses and quantitatively test their predictions in different scenarios. For example, an alternative explanation for behaviour in public goods games is that individuals are trying to maximise their income, but they are not perfectly behaving robots – instead, they start playing with a 'best guess' and then improve their play over time, by learning from how their behaviour influences their income (Burton-Chellew et al., 2015; Burton-Chellew & West, 2021; Nax et al., 2016). This is the imperfect or confused learners hypothesis.

When the explanatory behaviour of the prosocial preferences and imperfect learners hypothesis were tested quantitatively against one another, it was found that: (i) the imperfect learning hypothesis was better able to explain the data; and (ii) social preferences, if they exist, are more anti-social than pro-social(Burton-Chellew et al., 2015; Burton-Chellew & West, 2021, 2022; Nax et al., 2016). These same conclusions have been reached by individual experiments and by comparing the results across 237 public goods games (Burton-Chellew & West, 2021). Again, both the prosocial preferences hypothesis and the assumption that individuals were playing perfectly were rejected.

6.5. Different ways

Our aim here was not to provide a detailed assessment of the support for different competing hypotheses. Instead, we wanted to explore how the development of controls and null hypotheses, and the testing of competing hypotheses, can lead to different conclusions being reached. This is a basic tenet of science. In this case, the use of different controls or null hypotheses all suggested that the prosocial preferences hypothesis should be rejected. Further work could be designed to test between competing hypotheses – what is crucial is that there is not a bias towards certain hypotheses. Different experiments will have different pros and cons, and we need to look at what the body of data shows, across different experiments. Other important issues include the influence of variation in information, or the values of the game parameters, and why cooperation can briefly resume when the entire game is replayed (the 're-start' effect) (Burton-Chellew, 2022; Burton-Chellew et al., 2015; Burton-Chellew et al., 2017; Burton-Chellew & West, 2013; Camerer, 2013).

It is also important to apply the same criteria consistently, when competing different hypotheses. For example, it has been suggested that making individuals play into a black box or with computers is unnatural and so doesn't make sense (Camerer, 2013). However, this 'not natural' argument would similarly apply to many economic games. For example, strictly anonymous games played with no potential for repeated interactions are also unnatural when compared with most human interactions, and the conditions under which they have evolved in the real world. Consequently, the same logic that would exclude data from games with computers would exclude using economic games to quantitatively measure social preferences. We do not believe that laboratory experiments are useless (far from it!), just that care must be taken with their interpretation.

We also need to distinguish between experimental elaborations that test competing hypotheses, versus those that elaborate on a single

hypothesis. Returning to our hypothetical water colour experiment, we could envisage a whole field that examined more elaborate details but never addressed the fundamental problems with the original experiment. For example, how warmer water, or different shaped mugs (all made of ochre), lead to faster colour change, and how success in these experiments supported the original hypothesis that putting water into a drinking vessel turns it red. If experiments are not elaborated to explicitly test competing hypotheses, then entire fields could be built upon incorrect assumptions.

6.6. Response to potential problems

It can be hard for researchers working within a methodological paradigm to change their methods (Gelman & Loken, 2013, 2015). Consistent with this, the response in the public goods literature over the last two decades has been to not engage with these potential methodological problems. One approach has been to focus on post-hoc discussion of existing data, rather than designing ways to explicitly test between competing hypotheses (Camerer, 2013). But post-hoc explanations can often be constructed for multiple hypotheses (Burton-Chellew et al., 2015). Another approach has been to suggest that these experimental design issues represent something that "might prove useful in other species" (Camerer, 2013) or that they represent one of many potential "alternate theories" (Gächter & Fages, 2023). This is trying to sidestep a fundamental aspect of the scientific method that should apply to any form of experiment, across the natural and life sciences. In most cases, however, no mention of the potential problems are made, and everyone carries on as normal, just like the officials and townsfolk when they are faced with a naked emperor in *The Emperor's New Clothes* (Fig. 5) (Battu, 2023; Fehr & Charness, 2023; Fehr & Schurtenberger, 2018; Gächter, Kölle, & Quercia, 2017; Isler & Gächter, 2022; Weber et al., 2018).

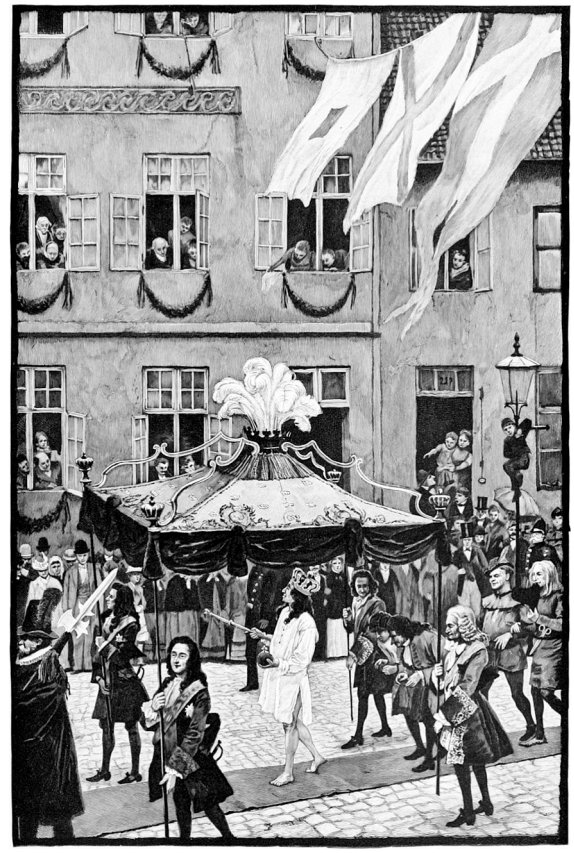

Fig. 5. The Emperor's New Clothes. A folktale by Hans Christian Andersen, in which conmen trick an emperor. The emperor has an obsession with fancy clothes and the conmen pose as weavers who will produce him a magnificent outfit that is invisible to incompetent or stupid people. The conmen pretend to produce the clothes and dress the emperor, who then sets off in a public procession. The emperor, his officials and the townsfolk all pretend that they can see clothes, to avoid appearing stupid. Eventually a child exclaims that the emperor is wearing nothing at all, and everyone realises that they have been fooled. Illustration by Hans Tegner in Andersen (1900).

7. Other examples

We have provided a detailed discussion of a single case, to illustrate the general problem. Nonetheless, even just within the wider public goods literature, there are other examples of analogous problems. For example, the development of controls to distinguish between the benefits to the individual and the group, of being in more cooperative groups (Jordan, Jordan, & Rand, 2017), or the need for experiments with more behavioural options, and less explicit prompts, when studying punishment (Pedersen, Kurzban, & McCullough, 2013; Pedersen, McAuliffe, & McCullough, 2018). Another example is the need for the development of control groups with no meaning when testing for ingroup biases (Dunham, 2018).

Another way of looking at the issues that we have raised is to distinguish between hypothesis testing and measurement. One approach has been to assume that behaviour in economic games perfectly represents "preferences for cooperation", and to then use data from such economic games to measure preferences (Gächter & Fages, 2023; Isler & Gächter, 2022). However, this should be seen as a hypothesis that needs to be tested with appropriate controls, not a starting assumption.

The measurement approach implicitly assumes that all other potentially confounding factors have been eliminated from an experiment, such that competing hypotheses are not possible. This can be very hard to do (impossible?), especially because there will often be potential confounding factors that the experimenter will not have thought of. It is precisely for this reason that control treatments are so useful in science (Ruxton & Colegrave, 2016).

Analogous examples can also be found outside of human behaviour. Genetic and ecological neutral theories have been proposed to explain molecular variation and species diversity (Bell, 2001; Hubbell, 2001; Kimura, 1983). The genetic neutral theory assumed that all genotypes have equal fitness, and the biodiversity neutral theory assumed that all species are equally equivalent. One potential problem with the genetic and ecological neutral theories was that it was so easy for the data to fit their null hypotheses (Leigh, 2007; Nee, 2005). Even when their null hypotheses were met, it would have been unwise to accept their assumptions that mutations do not affect fitness or that all tree species are the same. The neutral theories provide null models against which other theories can be tested (Bell, 2001; Kimura, 1983; Leigh, 2007; Leigh, Wright, Herre, & Putz, 1993; Nee, 2005).

8. Conclusions

A simple and elegant experiment, such as the standard public goods game, can be great for generating hypotheses. However, once a hypothesis has been generated, there is a need to develop competing hypotheses, and design new experiments that explicitly test between these competing hypotheses. Replication is good, but it useful to design different experiments that test the same hypothesis in different ways (e.g. with different control treatments), test competing hypotheses, or make a strong attempt to falsify the hypothesis (Nosek & Errington, 2020).

More generally, it is an exciting time for human behavioural studies, with methodological developments that are revolutionising the field. It is widely appreciated how previously methodologies were likely to generate false positive results, through mechanisms that have been termed "p-hacking", "fishing expeditions" or "forking path" analyses (Forstmeier, Wagenmakers, & Parker, 2016; Gelman & Loken, 2013; Higginson & Munafò, 2016; Ioannidis, 2005; Nissen et al., 2016;

Smaldino & McElreath, 2016). Furthermore, these problems can be compounded when experiments are not replicated, or if there is a publication bias towards certain results(Palmer, 2000). We add to this the need to repeat the testing of hypotheses, but with experiments that do this in different ways, allowing hypotheses a fair chance to be falsified or competing hypotheses to be tested (Nosek & Errington, 2020). Merely repeating the same treatment, even across different cultures, risks the canonization of unwarranted conclusions.

CRediT authorship contribution statement

Stuart A. West: Conceptualization, Writing – original draft. **Maxwell N. Burton-Chellew:** Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Stuart West reports financial support was provided by European Research Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank: J-B Andre, David Haig, Debra Lieberman, Ashley Thomas, Jevin West and especially Thom Scott-Phillips for extremely useful discussion / comments; the ERC (834164) for funding.

References

Andersen, H. C. (1900). *Fairy tales and stories*. New York: The Century Co.

Andreoni, J. (1995a). Cooperation in public-goods experiments: Kindness or confusion? *The American Economic Review, 85*(4), 891–904. https://doi.org/10.2307/2118238

Andreoni, J. (1995b). Warm-glow versus cold-prickle: The effects of positive and negative framing on cooperation in experiments. *The Quarterly Journal of Economics, 110*(1), 1–21.

Bateson, M., Nettle, D., & Roberts, G. (2006). Cues of being watched enhance cooperation in a real-world setting. *Biology Letters, 2*(3), 412–414. https://doi.org/10.1098/rsbl.2006.0509

Battu, B. (2023). Co-evolution of conditional cooperation and social norm. *Scientific Reports, 13*(1), 16625. https://doi.org/10.1038/s41598-023-43918-w

Bell, G. (2001). Neutral macroecology. *Science, 293*(5539), 2413–2418. https://doi.org/10.1126/science.293.5539.2413

Bergstrom, C. T., & West, J. D. (2021). *Calling bullshit: The art of skepticism in a data-driven world*. Random House.

Binmore, K., & Shaked, A. (2010). Experimental economics: Where next? *Journal of Economic Behavior & Organization, 73*(1), 87–100. https://doi.org/10.1016/j.jebo.2008.10.019

Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2021). *Introduction to meta-analysis* (2nd ed.). Wiley.

Burnham, T. C., & Hare, B. (2007). Engineering human cooperation. *Human Nature, 18*(2), 88–108. https://doi.org/10.1007/s12110-007-9012-2

Burton-Chellew, M. N. (2022). The restart effect in social dilemmas shows humans are self-interested not altruistic. *Proceedings of the National Academy of Sciences of the United States of America, 119*(49), Article e2210082119. https://doi.org/10.1073/pnas.2210082119

Burton-Chellew, M. N., & D'Amico, V. (2021). A preference to learn from successful rather than common behaviours in human social dilemmas. *Proceedings of the Royal Society B, 288*(1965), 20211590. https://doi.org/10.1098/rspb.2021.1590

Burton-Chellew, M. N., D'Amico, V., & Guérin, C. (2022). The strategy method risks conflating confusion with a social preference for conditional cooperation in public goods games. *Games, 13*(6), Article 69. https://doi.org/10.3390/g13060069

Burton-Chellew, M. N., & Guérin, C. (2022). Self-interested learning is more important than fair-minded conditional cooperation in public-goods games. *Evolutionary Human Sciences, 4*, Article e46. https://doi.org/10.1017/ehs.2022.45

Burton-Chellew, M. N., Mouden, C. E., & West, S. A. (2016). Conditional cooperation and confusion in public-goods experiments. *Proceedings of the National Academy of Sciences of the United States of America, 113*(5), 1291–1296. https://doi.org/10.1073/pnas.1509740113

Burton-Chellew, M. N., Mouden, C. E., & West, S. A. (2017). Social learning and the demise of costly cooperation in humans. *Proceedings of the Royal Society B: Biological Sciences, 284*(1853), Article 20170067. https://doi.org/10.1098/rspb.2017.0067

Burton-Chellew, M. N., Nax, H. H., & West, S. A. (2015). Payoff-based learning explains the decline in cooperation in public goods games. *Proceedings of the Royal Society B: Biological Sciences, 282*(1801), 20142678. https://doi.org/10.1098/rspb.2014.2678

Burton-Chellew, M. N., & West, S. A. (2013). Prosocial preferences do not explain human cooperation in public-goods games. *Proceedings of the National Academy of Sciences of the United States of America, 110*(1), 216–221. https://doi.org/10.1073/pnas.1210960110

Burton-Chellew, M. N., & West, S. A. (2021). Payoff-based learning best explains the rate of decline in cooperation across 237 public-goods games. *Nature Human Behaviour, 5*(10), 1330–1338. https://doi.org/10.1038/s41562-021-01107-7

Burton-Chellew, M. N., & West, S. A. (2022). The black box as a control for payoff-based learning in economic games. *Games, 13*(6), Article 76. https://doi.org/10.3390/g13060076

Camerer, C. F. (2013). Experimental, cultural, and neural evidence of deliberate prosociality. *Trends in Cognitive Sciences, 17*(3), 106–108. https://doi.org/10.1016/j.tics.2013.01.009

Camerer, C. F., Dreber, A., Holzmeister, F., Ho, T.-H., Huber, J., Johannesson, M., … Wu, H. (2018). Evaluating the replicability of social science experiments in nature and science between 2010 and 2015. *Nature Human Behaviour, 2*(9), 1–10. https://doi.org/10.1038/s41562-018-0399-z

Derksen, M., & Morawski, J. (2022). Kinds of replication: Examining the meanings of "conceptual replication" and "direct replication.". *Perspectives on Psychological Science, 17*(5), 1490–1505. https://doi.org/10.1177/17456916211041116

Dunham, Y. (2018). Mere Membership. *Trends in Cognitive Sciences, 22*(9), 780–793.

Fehr, E., & Charness, G. (2023). Social preferences: Fundamental characteristics and economic consequences. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4472932

Fehr, E., & Fischbacher, U. (2003). The nature of human altruism. *Nature, 425*(6960), 785–791. https://doi.org/10.1038/nature02043

Fehr, E., & Gächter, S. (2002). Altruistic punishment in humans. *Nature, 415*(6868), 137–140.

Fehr, E., & Schurtenberger, I. (2018). Normative foundations of human cooperation. *Nature Human Behaviour, 2*(7), 458–468. https://doi.org/10.1038/s41562-018-0385-5

Ferraro, P. J., & Vossler, C. A. (2017). The source and significance of confusion in public goods experiments. *Journal of the Economic Science Association, 10*(1), 53.

Fisher, R. A. (1935). *The design of experiments*. Macmillan.

Fletcher, S. C. (2021). The role of replication in psychological science. *European Journal for Philosophy of Science, 11*(1), Article 23. https://doi.org/10.1007/s13194-020-00329-2

Forstmeier, W., Wagenmakers, E.-J., & Parker, T. H. (2016). Detecting and avoiding likely false-positive findings – A practical guide. *Biological Reviews, 92*(2), 3–28. https://doi.org/10.1111/brv.12315

Gächter, S., & Fages, D. M. (2023). *Using the strategy method and beliefs to explain group size and MPCR effects in public good experiments*. Durham University. https://durham-repository.worktribe.com/output/1748683.

Gächter, S., Kölle, F., & Quercia, S. (2017). Reciprocity and the tragedies of maintaining and providing the commons. *Nature Human Behaviour, 1*(9), 650–656. https://doi.org/10.1038/s41562-017-0191-5

Gelman, A., & Loken, E. (2013). *The garden of forking paths: Why multiple comparisons can be a problem, even when there is no "fishing expedition" or "p-hacking" and the research hypothesis*. Unpublished Manuscript. Department of Statistics, Columbia University http://www.stat.columbia.edu/~gelman/research/unpublished/P_hacking.pdf.

Gelman, A., & Loken, E. (2015). The statistical crisis in science. *American Scientist, 103*(6), 460–465.

Gintis, H., Bowles, S., Boyd, R., & Fehr, E. (2003). Explaining altruistic behavior in humans. *Evolution and Human Behavior, 24*(3), 153–172. https://doi.org/10.1016/s1090-5138(02)00157-5

Hagen, E. H., & Hammerstein, P. (2006). Game theory and human evolution: A critique of some recent interpretations of experimental games. *Theoretical Population Biology, 69*(3), 339–348. https://doi.org/10.1016/j.tpb.2005.09.005

Haley, K. J., & Fessler, D. M. T. (2005). Nobody's watching? *Evolution and Human Behavior, 26*(3), 245–256. https://doi.org/10.1016/j.evolhumbehav.2005.01.002

Head, M. L., Holman, L., Lanfear, R., Kahn, A. T., & Jennions, M. D. (2015). The extent and consequences of P-hacking in science. *PLoS Biology, 13*(3), Article e1002106. https://doi.org/10.1371/journal.pbio.1002106

Hedges, L. V., & Schauer, J. M. (2021). The design of replication studies. *Journal of the Royal Statistical Society: Series A (Statistics in Society), 184*(3), 868–886. https://doi.org/10.1111/rssa.12688

Henrich, J., Boyd, R., Bowles, S., Camerer, C. F., Fehr, E., Gintis, H., … Tracer, D. (2005). "Economic man" in cross-cultural perspective: Behavioral experiments in 15 small-scale societies. *Behavioral and Brain Sciences, 28*(6), 795–815.

Higginson, A. D., & Munafò, M. R. (2016). Current incentives for scientists lead to underpowered studies with erroneous conclusions. *PLoS Biology, 14*(11), Article e2000995. https://doi.org/10.1371/journal.pbio.2000995

Houser, D., & Kurzban, R. (2008). Revisiting kindness and confusion in public goods experiments. *Journal of Public Economics, 92*(4), 1062–1069. https://doi.org/10.1016/j.jpubeco.2007.07.003

Hubbell, S. P. (2001). *The unified neutral theory of biodiversity and biogeography*. Princeton University Press.

Ioannidis, J. P. A. (2005). Why most published research findings are false. *PLoS Medicine, 2*(8), Article e124. https://doi.org/10.1371/journal.pmed.0020124

Isler, O., & Gächter, S. (2022). Conforming with peers in honesty and cooperation. *Journal of Economic Behavior & Organization, 195*, 75–86. https://doi.org/10.1016/j.jebo.2021.12.026

Jordan, M. R., Jordan, J. J., & Rand, D. G. (2017). No unique effect of intergroup competition on cooperation: Non-competitive thresholds are as effective as competitions between groups for increasing human cooperative behavior. *Evolution and Human Behavior, 38*(1), 102–108. https://doi.org/10.1016/j.evolhumbehav.2016.07.005

Kimura, M. (1983). *The neutral allele theory of molecular evolution*. Cambridge University Press.

Kümmerli, R., Burton-Chellew, M. N., Ross-Gillespie, A., & West, S. A. (2010). Resistance to extreme strategies, rather than prosocial preferences, can explain human cooperation in public goods games. *Proceedings of the National Academy of Sciences, 107*(22), 10125–10130. https://doi.org/10.1073/pnas.1000829107

Ledyard, J. O. (1995). Public goods: A survey of experimental research. In J. H. Kagel, & A. E. Roth (Eds.), *Handbook of experimental economics* (pp. 1–95). Princeton University Press.

Leigh, E. G. (2007). Neutral theory: A historical perspective. *Journal of Evolutionary Biology, 20*(6), 2075–2091. https://doi.org/10.1111/j.1420-9101.2007.01410.x

Leigh, E. G., Wright, S. J., Herre, E. A., & Putz, F. E. (1993). The decline of tree diversity on newly isolated tropical islands: A test of a null hypothesis and some implications. *Evolutionary Ecology, 7*(1), 76–102. https://doi.org/10.1007/bf01237735

Locey, M. L. (2020). The evolution of behavior analysis: Toward a replication crisis? *Perspectives on Behavior Science, 43*(4), 655–675. https://doi.org/10.1007/s40614-020-00264-w

Nax, H. H., Burton-Chellew, M. N., West, S. A., & Young, H. P. (2016). Learning in a black box. *Journal of Economic Behavior and Organization, 127*, 1–15. https://doi.org/10.1016/j.jebo.2016.04.006

Nee, S. (2005). The neutral theory of biodiversity: Do the numbers add up? *Functional Ecology, 19*(1), 173–176.

Nissen, S. B., Magidson, T., Gross, K., & Bergstrom, C. T. (2016). Publication bias and the canonization of false facts. *eLife, 5*, Article e21451. https://doi.org/10.7554/elife.21451.001

Nosek, B. A., & Errington, T. M. (2020). What is replication? *PLoS Biology, 18*(3), Article e3000691. https://doi.org/10.1371/journal.pbio.3000691

Nosek, B. A., Hardwicke, T. E., Moshontz, H., Allard, A., Corker, K. S., Dreber, A., … Vazire, S. (2022). Replicability, robustness, and reproducibility in psychological science. *Annual Review of Psychology, 73*, 1–30. https://doi.org/10.1146/annurev-psych-020821-114157

Palmer, A. R. (2000). Quasi-replication and the contract of error: Lessons from sex ratios, heritabilities and fluctuating asymmetry. *Annual Review of Ecology and Systematics, 31*, 441–480. https://doi.org/10.1146/annurev.ecolsys.31.1.441

Pedersen, E. J., Kurzban, R., & McCullough, M. E. (2013). Do humans really punish altruistically? A closer look. *Proceedings of the Royal Society B: Biological Sciences, 280*(1758), 20122723. https://doi.org/10.1098/rspb.2012.2723

Pedersen, E. J., McAuliffe, W. H. B., & McCullough, M. E. (2018). The unresponsive avenger: More evidence that disinterested third parties do not punish altruistically. *Journal of Experimental Psychology: General, 147*(4), 514–544. https://doi.org/10.1037/xge0000410

Popper, K. (1959). *The logic of scientific discovery*. Taylor & Francis.

Ruxton, G. D., & Colegrave, N. (2016). *Experimental design for the life sciences* (4th ed.). Oxford University Press.

Sefton, M., & Steinberg, R. (1996). Reward structures in public good experiments. *Journal of Public Economics, 61*(2), 263–287. https://doi.org/10.1016/0047-2727(95)01534-5

Shapiro, D. A. (2008). The role of utility interdependence in public good experiments. *International Journal of Game Theory, 38*(1), 81–106. https://doi.org/10.1007/s00182-008-0141-6

Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology. *Psychological Science, 22*(11), 1359–1366. https://doi.org/10.1177/0956797611417632

Smaldino, P. E., & McElreath, R. (2016). The natural selection of bad science. *Royal Society Open Science, 3*(9), Article 160384. https://doi.org/10.1098/rsos.160384

Thöni, C., & Volk, S. (2018). Conditional cooperation: Review and refinement. *Economics Letters, 171*, 37–40. https://doi.org/10.1016/j.econlet.2018.06.022

Weber, T. O., Weisel, O., & Gächter, S. (2018). Dispositional free riders do not free ride on punishment. *Nature Communications, 9*, 2390. https://doi.org/10.1038/s41467-018-04775-8

Zelmer, J. (2003). Linear public goods experiments: A meta-analysis. *Experimental Economics, 6*(3), 299–310. https://doi.org/10.1023/a:1026277420119